

Homework (Habanero)**Q1.** What is the standard form of a quadratic function?

- (A) $ax^2 + bx + c$
 (B) $ax + bx^2 + c$
 (C) $ax^2 - bx + c$
 (D) $ax^2 + b$
 (E) $ax + c$

Q2. Which of the following is a quadratic function in standard form?

- (A) $x^2 + 3x - 2$
 (B) $x^2 - 3x$
 (C) $x + 2$
 (D) $x^3 + 2x^2$
 (E) $x^2 - 3$

Q3. A doctor prescribes a patient 125 mg of a medication. The amount of medication in the patient's system after t hours is given by $M(t) = -2t^2 + 10t + 50$. What is the standard form of this function?

- (A) $-2t^2 + 10t + 50$
 (B) $2t^2 - 10t - 50$
 (C) $-t^2 + 5t + 25$
 (D) $t^2 - 5t - 25$
 (E) $t^2 + 5t + 25$

Q4. The growth of a cell population is modeled by $P(t) = 3t^2 + 2t + 10$, where t is time in hours. What is the key feature of this quadratic function?

- (A) Vertex at $(0, 10)$
 (B) Axis of symmetry at $t = -1$
 (C) Maximum value at $t = 2$
 (D) Minimum value at $t = -1$
 (E) None of these

Q5. Which of the following is a key feature of the quadratic function $f(x) = x^2 - 4x - 3$?

- (A) Vertex at $(2, -7)$
 (B) Axis of symmetry at $x = -2$
 (C) Maximum value at $x = -1$
 (D) Minimum value at $x = 2$
 (E) None of these

Q6. A quadratic function has its vertex at $(2, 3)$. The equation of this function in standard form is $f(x) = a(x - h)^2 + k$, where a , h , and k are constants. What is h ?

- (A) 0
 (B) 2
 (C) 3
 (D) -2
 (E) -3

Q7. A medical researcher is studying the effect of a new medication on patients. The dosage D of the medication is related to the number of patients P by $D(P) = 2P^2 - 3P + 1$. What is the standard form of this function?

- (A) $2P^2 - 3P + 1$
 (B) $P^2 - 2P - 3$
 (C) $-2P^2 + 3P - 1$
 (D) $-P^2 + 2P + 3$
 (E) $P^2 + 2P - 3$

Q8. Which of the following quadratic functions has a minimum value at $x = 1$?

- (A) $x^2 - 2x - 1$
 (B) $x^2 + 2x + 1$
 (C) $-x^2 + 2x - 1$
 (D) $-x^2 - 2x + 1$
 (E) $x^2 + 1$

Open Ended Questions (FRQ)**Q9.** The spread of a disease in a population is modeled by $S(t) = -t^2 + 10t + 20$, where t is time in days. Describe the key features of this quadratic function and sketch its graph.**Q10.** A quadratic function has its vertex at $(1, 4)$ and passes through the point $(2, 3)$. Find the equation of this function in standard form $f(x) = a(x - h)^2 + k$.**Chef's Tips**

- Remind students to show all work for free-response questions.
- Allow students to use graphing calculators to check their work.
- Encourage students to use real-world examples to illustrate key features of quadratic functions.